Uttarakhand-based Himalayan Environmental Studies and Conservation Organisation (HESCO). "Our road-building agencies need to learn from other countries on how to protect our mountains."

THE FRAGILE HILLS

The Char Dham project has planned two tunnels to avoid landslide-prone stretches, one of them being the 4.5 km Silkyara-Barkot route that saw the cave-in. In its final report to the Supreme Court, the high-powered committee had also recommended more tunnels in critically sensitive stretches. "The cost of tunnels can run into hundreds of crores, several times the cost of roads. The country cannot afford it everywhere," says minister Gadkari. The Indian Roads Congress, the apex body of highway engineers, has construction manuals for building weather-resilient roads in landslide-prone areas. In spite of this, experts find blunders on the ground. "The angle of hill cutting for road construction should be between 30 and 45 degrees in soil, 75 to 80 degrees in soft rock slopes, and 80 to 90 degree in hard rock slopes. Cutting more than these values or uncontrolled ballasting creates unstable slopes," says Anil Sinha, head of geotechnical engineering, Central Road Research Institute (CRRI).

In audits of problematic zones in Himalayan regions, the CRRI has pointed out several common "causative factors" behind hill roads collapsing, such as lack of sufficient drainage. "Due to the presence of water, rocks rich in clay minerals get saturated and lose their strength and become more vulnerable to weathering," says a CRRI report. Then there are "man-made reasons". "Vegetation cover is essential to check infiltration of surface water.... It has been observed that the soil, construction and demolition waste are dumped in the slide area which does not allow the vegetation to grow on the slopes," says one report from the Northeast.



"Roads are built in the hills because people there demand connectivity. It's a tough challenge but we are working on it"

NITIN GADKARI
Union minister, MoRTH

LOOKING ABROAD FOR SOLUTIONS

With losses mounting every year and the clamour for a permanent fix getting louder, the government has also decided to seek outside help. "Countries like Switzerland and Norway have worked on building climate-resilient roads in the mountains through innovative means," Union minister of road transport and highways Nitin Gadkari told INDIA TODAY. "We are exploring their engineering technology." The minister recently had experts over to demonstrate engineering solutions that have been successful in Switzerland and that could be replicated in the Himalayas. "Not enough work has happened in India on this [studying landslides and how they happen]. We are creating a separate division in road-building organisations to work on the problem," he says. Like accident-prone 'black spots' on highways, the minister says a map of landslide-prone spots will be made for all mountains across India.

"Where Switzerland has been innovative is in the adoption of a country-wide detailed hazard and risk assessment of all national roads," says Prof. Luuk Dorren, head of research and consultancy, Bern University of Applied Sciences. "This assessment forms the basis for defining natural hazard hotspots and, importantly, calculating the funds that can be invested for protective measures so that damage and resulting losses are in equilibrium," he says. Since the 1980s, only six motorists have died due to natural hazards on Swiss roads; the last incident was a rockfall that killed two in 2006.

A Swiss engineering firm that made a presentation to Gadkari has showcased technology tried and tested in the Alps. "We think the Alps are similar to the Himalayas in this regard," says a company official. Two kinds of protection have been suggested against landslides: active and passive. Active protection involves installing steel wire meshes on hill slopes to stop it from destabilising. Passive systems are like catching nets or physical covers over underlying infrastructure.

889 KM
Total length of highways

601 KM
of the project completed

CHAR DHAM PROJECT

The roads connect some of India's holiest shrines. Some routes also hold strategic value

ROADS



LENGTH

- 95 km
- 124 km
- 144 km
- 140 km
- 76 km
- 160 km
- 150 km

NON-STRATEGIC ROADS

Rudraprayag-Gaurikund and Dharasu Bend-Jankichatti

STRATEGIC ROADS

Rishikesh-Mana,
Rishikesh-Gangotri
Tanakpur-Pithoragarh